

Practical Workbook
CT-265
Introduction to Tools and Techniques for
Data Science



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S. No.	Object	Date
01	To build a strong foundation in Python programming by understanding the basic syntax, data types, operators, control flow, functions, and commonly used data structures.	12-9-2026
02	Learn how to use Pandas to make and work with Data Frames.	
03	To understand the NumPy library, which provides support for large, multi-dimensional arrays and matrices.	
04	To create effective static visualizations for data exploration and insight extraction.	
05	To apply foundational probability and statistics concepts to understand datasets.	
06	To provide a hands-on introduction to applying basic machine learning models for prediction.	
07	To acquire data from the web using web scraping techniques and extract structured information from websites.	
08	To apply natural language processing techniques to analyze and extract insights from customer review data.	
09	To understand the fundamentals of Microsoft Power BI for importing data, transforming it, building data models, creating visualizations, and designing interactive dashboards.	
10	To learn the basics of Tableau for connecting data, creating worksheets, applying filters, building dashboards, and presenting analytical stories from data.	

LAB # 01 OVERVIEW OF PYTHON

EXERCISE # 01: BASIC INPUT AND OUTPUT

Q1. Write a program that takes a student's name, roll number, and GPA as input and displays them clearly.

SOLUTION:

```
1  name=input("Enter your Name:")
2  rollno=int(input("Enter your Roll Number:"))
3  gpa=float(input("Enter your GPA:"))
4
5  print("Name:",name)
6  print("Roll No:",rollno)
7  print("GPA:",gpa)
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDs Labs\LAB#01> python Lab#01Ex#01Q1.py
Enter your Name:Moazzam Farooqui
Enter your Roll Number:68
Enter your GPA:3.84
Name: Moazzam Farooqui
Roll No: 68
GPA: 3.84
```

Q2. Create variables of type int, float, string, and bool, then print their values and data types using type()

SOLUTION:

```
1  rollno=68
2  gpa=3.84
3  name="Moazzam Farooqui"
4  is_student=True
5
6  print(rollno,type(rollno))
7  print(gpa,type(gpa))
8  print(name,type(name))
9  print(is_student,type(is_student))
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#01Q2.py
68 <class 'int'>
3.84 <class 'float'>
Moazzam Farooqui <class 'str'>
True <class 'bool'>
```

Q3. Take a number as input and print its square, cube, and half value.

SOLUTION:

```
1  number=float(input("Enter the number:"))
2
3  is_square=number*number
4  is_cube=number*number*number
5  is_half=number/2
6
7  print("Square of the number is:",is_square)
8  print("Cube of the number is:",is_cube)
9  print("Half of the number is:",is_half)
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#01Q3.py
Enter the number:5
Square of the number is: 25.0
Cube of the number is: 125.0
Half of the number is: 2.5
```

EXERCISE # 02: CONDITIONS AND LOOPS

Q1. Write a program to check whether a number is even or odd.

SOLUTION:

```
1  def is_even_odd():
2      number=int(input("Enter the number:"))
3      if(number%2==0):
4          print(f"{number} is an even number")
5      else:
6          print(f"{number} is an odd number")
7
8  is_even_odd()
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#02Q1.py
Enter the number:5
5 is an odd number
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#02Q1.py
Enter the number:6
6 is an even number
```

Q2. Print all numbers from 1 to 20 using a for loop.

SOLUTION:

```
1  for number in range(1,21):
2      print(number)
3
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#02Q2.py
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
```

Q3. Use a while loop to print the multiplication table of a given number.

SOLUTION:

```
1  number=int(input("Enter the number: "))
2
3  m=1
4
5  while m<=10:
6      print(f"{number}x{m}={number*m}")
7      m+=1
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#02Q3.py
Enter the number: 9
9x1=9
9x2=18
9x3=27
9x4=36
9x5=45
9x6=54
9x7=63
9x8=72
9x9=81
9x10=90
```

Q4. Challenge: Count how many numbers between 1 and 50 are divisible by 3.

SOLUTION:

```
1 count=0
2 for number in range(1,51):
3     if number%3==0:
4         count+=1
5
6 print(f"Numbers divisible by 3: {count}")
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#02Q4.py
Numbers divisible by 3: 16
```

EXERCISE # 03: FUNCTIONS AND DATA STRUCTURES

Q1. Create a function that accepts two numbers and returns the larger one.

SOLUTION:

```
1  def greater_number():
2      number1=int(input("Enter number 1:"))
3      number2=int(input("Enter number 2:"))
4      if(number1>number2):
5          return number1
6      else:
7          return number2
8
9  print(greater_number())
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#03q1.py
Enter number 1:4
Enter number 2:3
4
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#03q1.py
Enter number 1:6
Enter number 2:7
7
```

Q2. Store five course names in a list and print them one by one.

SOLUTION:

```
1  mylist=['DSA','DBMS','AIES','SE','ITTDS']
2
3  for courses in mylist:
4      print(courses)
```


OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#03Q2.py
DSA
DBMS
AIES
SE
ITTDS
```

Q3. Create a dictionary containing student information such as name, age, department, and semester.

SOLUTION:

```
1 mydictionary={
2     "Name":"Moazzam Farooqui",
3     "Age":19,
4     "Department":"CSIT",
5     "Semester":"5th"
6 }
7
8 print(mydictionary)
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#03Q3.py
{'Name': 'Moazzam Farooqui', 'Age': 19, 'Department': 'CSIT', 'Semester': '5th'}
```

Q4. Challenge: Write a function that accepts a list of numbers and returns the maximum, minimum, and average.

SOLUTION:

```
1  def calculate(numbers):
2      maximum=max(numbers)
3      minimum=min(numbers)
4      average=sum(numbers)/len(numbers)
5      return maximum,minimum,average
6
7  numbers=[1,2,3,4,5]
8
9  maximum,minimum,average=calculate(numbers)
10
11  print("Maximum:",maximum)
12  print("Minimum:",minimum)
13  print("Average:",average)
```

OUTPUT:

```
PS C:\users\user\Desktop\ITTDS Labs\LAB#01> python Lab#01Ex#03Q4.py
Maximum: 5
Minimum: 1
Average: 3.0
```